ROI-LGP: An ROI-Oriented Attack Against YOLO-Based Palmprint ROI Extraction

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Abstract. Palmprint Region of Interest (ROI) extraction is a critical preprocessing step in palmprint recognition, yet adversarial attacks against this stage remain underexplored. When directly applied to ROI extraction, however, existing object detection attacks overlook redundant candidates and the geometric dependencies required for ROI construction. In this paper, we propose ROI-LGP, an ROI-oriented adversarial attack against YOLO-based palmprint ROI extraction. Specifically, a Candidate Collaborative Suppression (CCS) module is constructed to jointly suppresses redundant candidates that can sustain ROI construction. An ROI-Sensitive Perturbation Allocation (RSPA) module is introduced to direct gradients toward task-relevant regions. Across YOLOv8m, YOLOv9m, and YOLOv11m, ROI-LGP consistently degrades detection performance and disrupts ROI extraction. On YOLOv11m, it achieves the highest ASR among the compared attacks with 94.02%, while maintaining a PSNR of 49.38 dB. Adequate experiments show that ROI-LGP effectively disrupts palmprint ROI extraction while preserving visual quality.

Keywords: Palmprint recognition · ROI extraction · adversarial attack

1 Introduction

Palmprint recognition has become an important research topic in the field of biometric identification due to its rich texture information, convenient acquisition, and high user acceptability [1] [2]. In a typical palmprint recognition pipeline, Region of Interest (ROI) extraction is a key preprocessing step that directly affects the downstream tasks of feature extraction and matching. But existing adversarial attacks on palmprint recognition [3] [4] mainly focus on the recognition model or matching stage, seeking to disrupt identity classification or feature matching. Consequently, the adversarial vulnerability of ROI extraction stage remains unexplored.

Many extraction methods localize the palm region using geometric structures such as finger-gap points, palm centers, and palm boundaries [5–7]. These

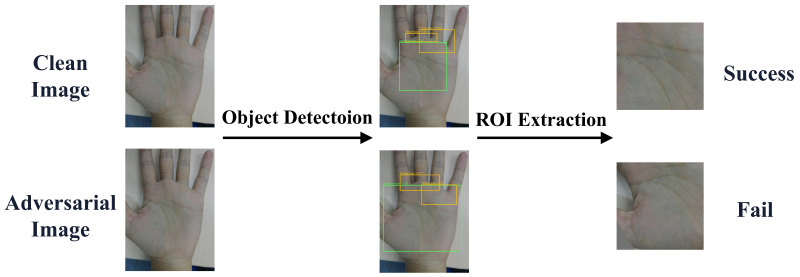


Fig. 1. Illustration of ROI extraction from clean and adversarial images.

structures determine the position, scale, and orientation of ROI, enabling the extraction of a normalized palmprint region. Because ROI construction depends on the spatial relationships among these structures, corrupting the detector outputs can cause ROI extraction to fail, as illustrated in Fig. 1.

However, directly applying existing object detection attacks to this process presents two limitations. First, detectors often produce multiple candidates around the same geometric structure. Suppressing the dominant candidates does not necessarily eliminate all usable detections. Neighboring or residual candidates may survive post-processing and still support ROI construction. Second, digital-domain attacks such as Targeted Objectness Gradient (TOG) [8] and Class-Wise Attack (CWA) [9] often optimize perturbations over the entire image without explicitly prioritizing ROI-sensitive areas.

To address these challenges, we develop ROI-LGP, an ROI-oriented adversarial attack framework built upon local perturbations with adaptively global attacks (LGP) [10]. ROI-LGP introduces Candidate Collaborative Suppression (CCS) to prevent critical geometric structures from being recovered through redundant detector responses. Rather than suppressing only dominant candidates, CCS also targets neighboring responses and alternative candidates that may survive or become selected during post-processing. We further introduce ROI-Sensitive Perturbation Allocation (RSPA) to align the spatial distribution of perturbations with the task-specific sensitivity of ROI construction. RSPA derives a spatial weight map from key regions, including finger-valley points and palm centers. This map guides perturbations toward regions closely associated with ROI construction while reducing unnecessary changes elsewhere. By jointly suppressing redundant candidates and reallocating perturbations toward ROI-sensitive regions, ROI-LGP effectively disrupts downstream ROI construction while preserving the perceptual quality of adversarial images.

The main contributions of this paper are summarized as follows:

1. We investigate adversarial attacks at the palmprint ROI extraction stage and propose ROI-LGP, which disrupts ROI construction by corrupting detections of key geometric primitives.

2. CCS is designed to jointly suppress dominant, neighboring and residual candidates, reducing the likelihood that usable key structures survive post-processing.
3. We introduce RSPA to guide perturbations toward ROI-sensitive regions through a spatial weight map, improving perturbation efficiency while preserving visual quality.

2 Related Work

2.1 Palmprint ROI Extraction

Palmprint ROI extraction aims to obtain a stable and normalized palm region from a palm image for subsequent recognition. Early methods typically segmented the hand region, located finger-gap points or boundary landmarks, and established a local coordinate system to extract a fixed-size ROI [11]. Recent methods have increasingly formulated ROI extraction as a detection or keypoint localization problem. Lighter Deep-learning-based Mobile Palmprint Verification (DeepMPV+) [12] detects finger-gap points and palm-center point as geometric primitives and constructs the palmprint ROI according to their spatial relationship. Luo and Zhong [7] proposed a robust and adaptive method that combines palm region detection and point localization for ROI extraction under unconstrained conditions. More recently, Grosz *et al.* [13] estimated nine keypoints along the palmar boundary and localized the ROI by mapping them to predefined destination points through a homographic transformation. Despite their different implementations, these methods share the principle of constructing normalized ROIs from localized points and their spatial relationships.

2.2 Adversarial Attacks on Object Detection

Adversarial attacks on object detection have targeted different components of the detection pipeline. Robust Adversarial Perturbation (R-AP) [14] and TOG [8] demonstrate that object detectors can be disrupted by attacking proposals, localization, class responses or objectness scores. Subsequent studies have further extended the attack surface. Phantom Sponges [15] targets non-maximum suppression, showing that detector post-processing is also vulnerable to adversarial attacks. Global Layout Aware Attacks on Object Detection (GLOW) [16] incorporates global layout information to attack object detectors in multi-object scenes. LGP [10] is particularly relevant to this work because it performs attacks at the proposal level. By jointly optimizing shape, location, and class losses, LGP can perturb the detection outputs used to localize ROI-related points. However, these methods are designed for general object detection and do not account for redundant candidate proposals or the spatial sensitivity of palmprint ROI construction.

3 Proposed Method

3.1 Overall Framework

Our method attacks the YOLO-based keypoint detector in the DeepMPV+ ROI extraction pipeline [12] . The detector represents each finger-gap or palm-center point as a bounding box, whose center defines the corresponding point location for ROI construction. As illustrated in Fig. 2, ROI-LGP integrates two task-oriented modules, CCS and RSPA. CCS suppresses dominant candidate proposals together with neighboring and residual ones that may remain effective after post-processing. RSPA applies a spatial weight map to perturbation gradients, emphasizing ROI-sensitive regions while preserving visual quality.

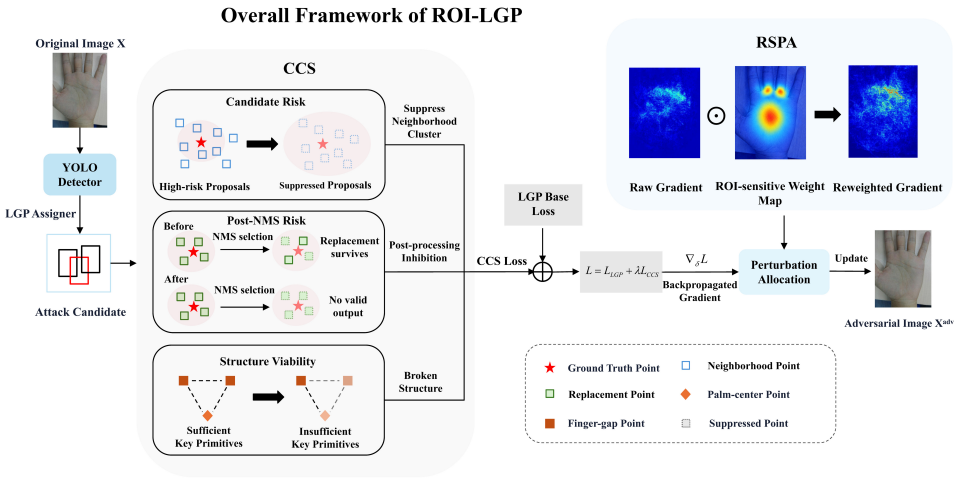


Fig. 2. Overall Framework of ROI-LGP

3.2 Candidate Collaborative Suppression

LGP [10] attacks object detectors by tracking and suppressing high-quality pre-NMS proposals associated with individual targets. In the DeepMPV+ pipeline, however, ROI construction remains feasible if post-processing retains two finger-gap points and one palm-center point. We therefore introduce CCS to jointly suppress proposals that can preserve this required point configuration. CCS models candidate risk, structure viability, and post-processing residual risk.

Let $\{(g_j, t_j)\}_{j=1}^{M}$ denote the annotated points, where g_j is the bounding-box representation of point j and t_j is its class label. Given the current pre-NMS proposals $P = \{(b_i, \mathbf{s}_i)\}_{i=1}^{N}$, we compute the overlap $o_{ij} = \mathrm{IoU}(b_i, g_j)$. The

candidate set is defined as $\mathcal{C}_j = \{i \mid o_{ij} > \tau_r\}$. If $\mathcal{C}_j$ is empty, the top-K proposals with the largest overlaps are used instead.

The candidate risk is measured by $r_{ij} = s_{i,t_j} o_{ij}$, where s_{i,t_j} is the target-class score of proposal b_i. For each annotated point, the top-K proposals with the largest candidate risks form the selected candidate set $\mathcal{K}_j$. The candidate risk loss is defined as

$$\mathcal{L}_{\text{risk}} = \frac{1}{|\mathcal{J}_r|} \sum_{j \in \mathcal{J}_r} \frac{1}{|\mathcal{K}_j|} \sum_{i \in \mathcal{K}_j} -\log\left(1 - \hat{s}_{i,t_j} + \epsilon\right) \left(0.5 + \text{sg}(o_{ij})\right) \ , \tag{1}$$

where $\hat{s}_{i,t_j}$ is the clipped target-class score, $\mathcal{J}_r$ contains points with valid selected candidates, and $\text{sg}(\cdot)$ denotes stop-gradient. Minimizing $\mathcal{L}_{\text{risk}}$ suppresses multiple high-risk proposals associated with each annotated point.

To model structure viability, the survival score of each point is defined as

$$v_j = \text{smax}_\tau \left(\{r_{ij} \mid i \in \mathcal{K}_j\}\right) \ , \tag{2}$$

where $\text{smax}_\tau(\cdot)$ denotes the smooth maximum with temperature τ. Let $v_{\text{fg}}^{(2)}$ denote the second-largest survival score among the finger-gap points. Let v_{pc} denote the smooth maximum survival score among the palm-center points. The structure-aware loss is formulated as

$$\mathcal{L}_{\text{struct}} = \text{smin}_\tau \left(\{v_{\text{fg}}^{(2)}, v_{\text{pc}}\}\right) \ , \tag{3}$$

where $\text{smin}_\tau(\cdot)$ denotes the smooth minimum with temperature τ. This loss measures whether the point configuration required for ROI construction remains viable.

To model post-processing residual risk, we introduce a center prior:

$$c_{ij} = \exp\left(-\frac{d_{ij}^2}{0.25}\right), \qquad d_{ij} = \frac{\|\mathbf{p}_i - \mathbf{p}_j\|_2}{D_j} \ , \tag{4}$$

where $\mathbf{p}_i$ and $\mathbf{p}_j$ are the centers of b_i and g_j, respectively, and D_j is the diagonal length of g_j. The residual risk is defined as $q_{ij} = s_{i,t_j} o_{ij}(1 + \eta c_{ij})$, where η controls the contribution of the center prior. For each annotated point, the top-K candidates with the largest residual risks form the residual candidate set $\mathcal{Q}_j$. The post-processing residual loss is defined as

$$\mathcal{L}_{\text{post}} = \frac{1}{|\mathcal{J}_p|} \sum_{j \in \mathcal{J}_p} \frac{1}{|\mathcal{Q}_j|} \sum_{i \in \mathcal{Q}_j} -\log\left(1 - \hat{s}_{i,t_j} + \epsilon\right) \left(0.5 + \text{sg}(o_{ij})\right) \left(1 + \eta\, \text{sg}(c_{ij})\right) \ , \tag{5}$$

where $\mathcal{J}_p$ contains points with valid residual candidate sets.

The overall CCS objective combines the three suppression terms:

$$\mathcal{L}_{\text{CCS}} = \lambda_{\text{risk}} \mathcal{L}_{\text{risk}} + \lambda_{\text{struct}} \mathcal{L}_{\text{struct}} + \lambda_{\text{post}} \mathcal{L}_{\text{post}} \ , \tag{6}$$

where λ_{risk}, λ_{struct}, and λ_{post} control the contributions of the three terms. By jointly suppressing high-risk candidates, the required point structure, and potential residual candidates, CCS adapts proposal-level attacks to the geometric dependencies of ROI construction.

3.3 ROI-Sensitive Perturbation Allocation

CCS suppresses redundant point-associated proposals at the proposal level. However, palmprint ROI extraction is also sensitive to the spatial distribution of perturbation gradients. In the considered pipeline, the finger-gap points and palm-center point determine the position, orientation, and scale of the extracted ROI. Perturbations around these points are therefore more likely to affect downstream ROI construction. RSPA addresses this issue by reweighting perturbation gradients according to ROI-sensitive regions.

Given an annotated point g_j, we denote its center, width, and height as (x_j, y_j), w_j, and h_j, respectively. For each pixel location (u, v), an elliptic Gaussian weight is defined as

$$G_j(u,v) = \exp\left(-\frac{1}{2}\left[\left(\frac{u-x_j}{\sigma w_j}\right)^2 + \left(\frac{v-y_j}{\sigma h_j}\right)^2\right]\right) , \tag{7}$$

where σ controls the spatial extent of the point-sensitive region.

The task-aware weight map is initialized with the background weight ω_b. For each point, its local weight is computed as

$$M_j(u,v) = \omega_b + \left(\omega_{t_j} - \omega_b\right) G_j(u,v) , \tag{8}$$

where ω_{t_j} denotes the class-specific peak weight associated with point g_j. In our implementation, the palm-center point is assigned a larger peak weight than the finger-gap points. The image-level weight map is obtained by retaining the maximum response across all point-sensitive regions:

$$M(u,v) = \max\left(\omega_b, \max_j M_j(u,v)\right) . \tag{9}$$

For the palm-center point, we further introduce a broader core region to cover the central palm area involved in ROI construction:

$$M(u,v) = \max\left(M(u,v), \omega_b + (\omega_c - \omega_b)\, G_c(u,v)\right) , \tag{10}$$

where $G_c(u, v)$ is computed using a larger spatial scale, and ω_c denotes the peak weight of the ROI core region.

RSPA introduces no additional loss term. Instead, it directly reweights the gradient used to update the perturbation. Let $\mathcal{L}_{\text{all}}$ denote the complete attack objective before gradient reweighting. At iteration t, the perturbation gradient is reweighted as

$$\widetilde{\nabla}_\delta^{(t)} = M^{(t)} \odot \nabla_\delta \mathcal{L}_{\text{all}} , \tag{11}$$

where $\odot$ denotes element-wise multiplication. The weight map $M^{(t)}$ is activated after a warm-up stage and gradually increased to its final strength. RSPA therefore leaves the attack objective unchanged while directing perturbation gradients toward regions that are more relevant to palm ROI construction.

4 Experiments

4.1 Dataset and Implementation

Dataset. We conduct experiments on the public Mobile Palmprint Database (MPD) introduced in DeepMPV+ [12], which contains 16,000 images from 200 identities. The identities are partitioned into disjoint training, validation and test sets at a ratio of 7:1:2. Each image is annotated with two finger-gap points and one palm-center point in the YOLO bounding-box format.

Implementation Details. We attack the DeepMPV+ ROI extraction pipeline using YOLO-based keypoint detectors. All input images are resized to 640×640 pixels. The perturbation is constrained by $\epsilon = 8/255$ and optimized for 50 iterations using Adamax with a learning rate of 0.003. For CCS, the candidate and post-processing branches retain 20 and 30 proposals respectively. RSPA assigns weights of 0.45, 1.0, 1.8, and 1.4 to the background, finger-gap, palm-center, and ROI-core regions respectively. All compared methods use the same perturbation budget and evaluation pipeline.

Evaluation Metrics. We evaluate attack performance in terms of detector degradation, attack success and visual quality. The drop in mAP50–95, calculated between clean and adversarial detection results, measures the degradation of YOLO keypoint detector. Attack success rate (ASR) is computed over samples accepted under clean conditions, where an attack succeeds if ROI extraction fails or the adversarial ROI fails downstream palmprint authentication. PSNR measures the visual similarity between clean and adversarial images. Higher mAP50–95 drops and ASR indicate stronger attack effectiveness, whereas higher PSNR indicates better visual quality.

4.2 Overall Attack Performance

We evaluate ROI-LGP on YOLOv8m, YOLOv9m, and YOLOv11m under the same perturbation budget and evaluation settings. All three detectors achieve an mAP50–95 of 0.931 on clean images. Table 1 reports detector performance degradation, attack success, and visual quality. It can be observed that ROI-LGP causes substantial mAP degradation across three YOLO detectors. Our method consistently achieves ASRs of at least 93.80% and a maximum of 94.57% on YOLOv9m. Meanwhile, PSNR remains above 49.3 dB, indicating that strong pipeline disruption is achieved while preserving visual quality. Figure 3 further illustrates how ROI-LGP disrupts the detection of finger-gap and palm-center structures in adversarial images.

4.3 Comparison with Existing Attacks

We compare ROI-LGP with representative object detection attacks, including TOG [8], CWA [9], Attention-Focused Offensive Gradient (AFOG) [17] and Customizable and Natural Camouflage Attack (CNCA) [18]. All methods are evaluated on YOLOv11m using the same test set and DeepMPV+ evaluation pipeline under an ℓ_∞ perturbation budget of $\epsilon = 8/255$.

Table 1. Overall attack performance on different YOLO detectors.

Detector	mAP50–95 drop	ASR (%)	PSNR
YOLOv8m	0.861	93.80	49.61
YOLOv9m	0.846	94.57	49.81
YOLOv11m	0.894	94.02	49.38

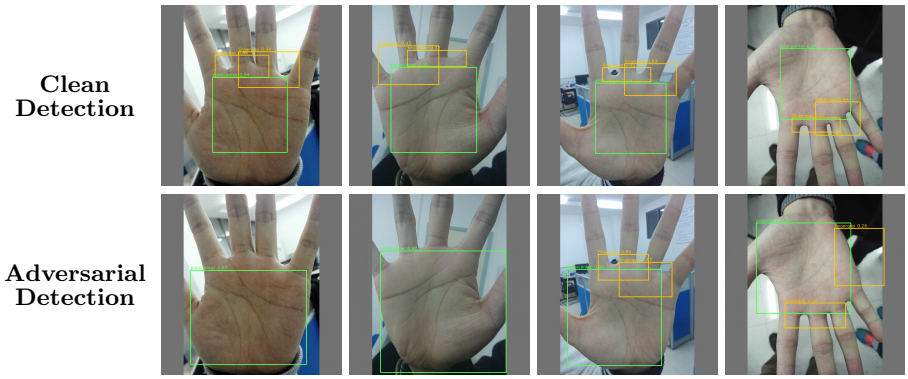


Fig. 3. Detection results on clean and adversarial palmprint images.

Table 2. Comparison with existing adversarial attacks on YOLOv11m. Best results are shown in bold and second-best results are underlined.

Method	mAP50–95 drop	ASR (%)	PSNR
TOG	<u>0.901</u>	92.64	43.33
CWA	0.307	90.72	<u>46.83</u>
AFOG	**0.916**	<u>93.53</u>	42.94
CNCA	0.572	54.12	31.01
Ours	0.894	**94.02**	**49.38**

As shown in Table 2, ROI-LGP achieves the highest ASR of 94.02% while maintaining a PSNR of 49.38 dB. Although TOG and AFOG cause higher degradation in mAP50–95, their ASRs are lower and their PSNR values are 43.33 and 42.94 dB respectively. This suggests that aggressively degrading detector outputs does not necessarily translate into more effective ROI disruption. The substantially lower ASR of LGP further indicates that its generic proposal-level objective does not fully capture the structural dependencies of palmprint ROI construction. By jointly suppressing redundant candidates and directing perturbations toward ROI-sensitive regions, ROI-LGP achieves a better balance between attack effectiveness and visual quality.

4.4 Ablation Study

To assess the contributions of CCS and RSPA, we add them individually and jointly to the LGP baseline and evaluate all variants on YOLOv11m under the

same settings. As shown in Table 3, LGP alone achieves an ASR of 27.06%, indicating limited effectiveness against ROI extraction. Adding CCS raises the ASR to 92.89%, providing the largest improvement among the individual modules. RSPA alone also increases the ASR to 54.72% while maintaining a PSNR of 50.71 dB. Combining CCS and RSPA achieves the highest ASR of 94.02% with a PSNR of 49.38 dB. These results show that CCS provides the primary improvement by addressing candidate redundancy and structural viability, while RSPA offers a complementary gain through spatial perturbation allocation.

Table 3. Ablation study of the proposed modules on YOLOv11m.

Method	+CCS	+RSPA	mAP50–95 drop	ASR (%)	PSNR
LGP			0.517	27.06	51.78
LGP	✓		0.890	92.89	49.38
LGP		✓	0.694	54.72	50.71
ROI-LGP	✓	✓	0.894	94.02	49.38

5 Conclusion

In this paper, we investigate adversarial attacks against palmprint ROI extraction and propose ROI-LGP for the YOLO-based DeepMPV+ pipeline. ROI-LGP uses CCS to suppress redundant candidates and RSPA to direct perturbations toward ROI-sensitive regions. Experiments on three YOLO detectors suggest that ROI-LGP consistently disrupts ROI extraction while preserving visual quality. Comparisons with existing attacks and ablation studies further support the effectiveness and complementary roles of CCS and RSPA. Currently, this study mainly evaluates YOLO-series detectors within a keypoint-detection-based ROI extraction pipeline. Future work will extend the attack framework to other detector architectures and different types of palm ROI extraction methods.

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